

PRACTICAL 6

PRACTICAL TITLE :- JSON Data and Fetch API

Problem Definition	Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.
Key Questions / Analysis / Interpretation	1. How is JSON fetched, parsed, and rendered? 2. Which array methods are used for search, filter, sort, and pagination? 3. How are loading and error states handled? 4. How is modularity maintained in JavaScript files?
Key Skills to be Addressed	JSON, Fetch API, arrays, map/filter/sort, modular JavaScript, pagination logic
Applications	Dynamic data rendering in modern web applications
Learning Outcome	Students will consume external JSON data and render reusable frontend views.
CO Mapping	CO3, CO4
Dataset/Test Data	events.json, students.json, faqs.json with at least 15 records each
Tools/Technology to be Used	JavaScript ES6+, JSON, HTML5, CSS3
Total Hours	Implementation: 4 hours Total engagement: 6 hours
Post Laboratory Work Description	Submit JSON files, rendered views, and test screenshots for search/filter/pagination.
Evaluation Strategy Including Viva	JSON parsing, array logic, UI rendering, error handling, and console testing.
Advanced/Intermediate Extension	Intermediate: Add dependent dropdowns for country/state/city. Advanced: Cache the latest JSON response in localStorage for offline-like display.

GIT HUB REPOSITORY :-

<https://github.com/nehvala0005-cloud/sh.git>

1. Aim

To fetch external JSON data using the Fetch API and dynamically display it on a webpage with search, filtering, sorting, and pagination functionality.

2. Problem Definition

Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

3. Key Questions / Analysis

Write these:

- 1. How is JSON fetched, parsed, and rendered?**
 - 2. Which array methods are used for search, filter, sort, and pagination?**
 - 3. How are loading and error states handled?**
 - 4. How is modularity maintained in JavaScript files?**
-

4. Key Skills to be Addressed

- JSON**
 - Fetch API**
 - JavaScript Arrays**
 - map()**
 - filter()**
 - sort()**
 - Modular JavaScript**
 - Pagination Logic**
 - Dynamic DOM Rendering**
-

5. Application

Dynamic data rendering in modern web applications.

6. Theory

JSON

JSON stands for JavaScript Object Notation. It is a lightweight format used to store and exchange structured data.

Example:

```
{  
  "title": "Web Development Workshop",  
  "category": "Workshop",  
  "date": "25 September 2026"  
}
```

Fetch API

The Fetch API is used to request data from external files or APIs.

In our StudentHub project, JavaScript fetches data from:

`../JSON/events.json`

The JSON response is then converted into JavaScript objects and displayed dynamically.

Array Methods

`filter()` → filters events according to a condition.

`map()` → creates HTML elements/data representation from event objects.

`sort()` → arranges data according to a particular order.

`slice()` → selects the records required for the current page.

7. Implementation

In the StudentHub project:

1. Event data is stored in `events.json`.
2. JavaScript uses the Fetch API to load the JSON file.
3. JSON data is converted into JavaScript objects.

4. Events are dynamically rendered on the webpage.
 5. Search is used to find events.
 6. Category filter is used to display selected event types.
 7. Pagination displays a limited number of events per page.
 8. Next and Previous buttons are used to navigate between pages.
-

8. Important Code

Practical file me apne actual events.js ka relevant code paste kar.

For example:

```
fetch("../JSON/events.json")
  .then(response => response.json())
  .then(data => {
    events = data;
    displayEvents();
  })
  .catch(error => {
    console.error("Error loading events:", error);
  });
```

Dynamic Rendering

```
function displayEvents() {
  eventContainer.innerHTML = "";

  currentEvents.forEach(event => {
    const card = document.createElement("div");

    card.innerHTML = `
```

```

        <h3>${event.title}</h3>
        <p>${event.date}</p>
        <p>${event.category}</p>
    `;

    eventContainer.appendChild(card);
});
}

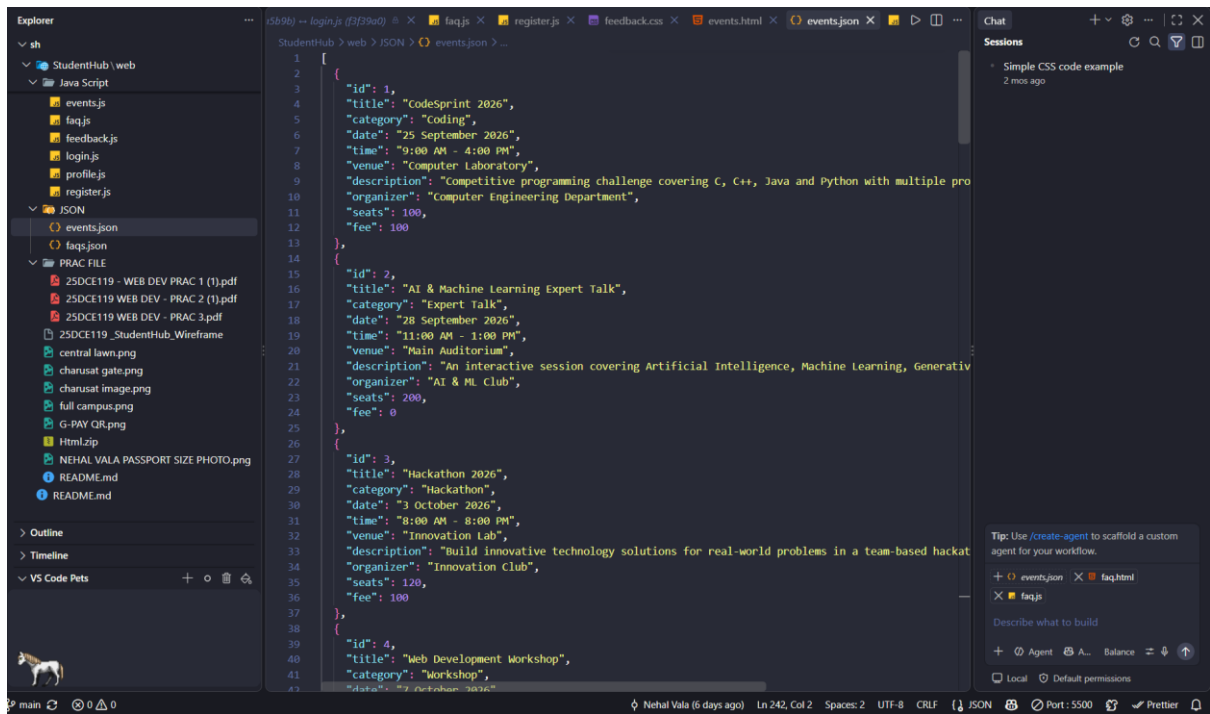
```

9. TEST CASES

Test Case Input/Action		Expected Result
1	Load Events page	Events fetched from JSON
2	Search event name	Matching events displayed
3	Select category	Relevant events displayed
4	Click Next	Next page displayed
5	Click Previous	Previous page displayed
6	Invalid/unavailable data	Error handled appropriately

PROOF OF WORK :-

6.1 :- JSON File



6.2 :- Events Dynamically Displayed



StudentHub Portal

Empowering Students Through Technology

[Home](#)
[About](#)
[Register](#)
[Login](#)
[Dashboard](#)
[Events](#)
[Profile](#)
[Contact](#)
[Admin](#)
[FAQ](#)
[Feedback](#)

Upcoming Events

Explore coding competitions, hackathons, workshops, expert talks, seminars and cultural activities happening on campus.

All Categories

Coding

CodeSprint 2026

Date: 25 September 2026
Time: 9:00 AM - 4:00 PM
Venue: Computer Laboratory
Organizer: Computer Engineering Department
Seats: 100
Competitive programming challenge covering C, C++, Java and Python with multiple problem-solving rounds.

₹100

Register

Expert Talk

AI & Machine Learning Expert Talk

Date: 28 September 2026
Time: 11:00 AM - 1:00 PM
Venue: Main Auditorium
Organizer: AI & ML Club
Seats: 200
An interactive session covering Artificial Intelligence, Machine Learning, Generative AI and future career opportunities.

Free

Register

Hackathon

Hackathon 2026

Date: 3 October 2026
Time: 8:00 AM - 8:00 PM
Venue: Innovation Lab
Organizer: Innovation Club
Seats: 120
Build innovative technology solutions for real-world problems in a team-based hackathon.

₹100

Register

Workshop

Web Development Workshop

Date: 7 October 2026
Time: 10:00 AM - 1:00 PM
Venue: Web Technology Lab
Organizer: Web Development Club
Seats: 80
Learn HTML, CSS, responsive design and modern web development through practical activities.

Free

Register

Seminar

Cyber Security Awareness Seminar

Date: 10 October 2026
Time: 2:00 PM - 4:00 PM
Venue: Seminar Hall
Organizer: Cyber Security Club
Seats: 150
Understand cyber threats, secure passwords, phishing, online privacy and basic security practices.

Free

Register

Coding

Java Programming Challenge

Date: 15 October 2026
Time: 9:30 AM - 2:00 PM
Venue: Programming Lab
Organizer: Programming Club
Seats: 100
Test your Java programming and problem-solving skills through a series of coding challenges.

₹100

Register

Expert Talk

Cloud Computing Expert Session

Date: 18 October 2026
Time: 11:00 AM - 12:30 PM
Venue: Main Auditorium
Organizer: Cloud Computing Club
Seats: 180
Explore cloud computing concepts, modern cloud platforms and career opportunities in cloud technology.

Free

Register

Workshop

Startup & Innovation Bootcamp

Date: 22 October 2026
Time: 10:00 AM - 4:00 PM
Venue: Innovation Center
Organizer: Entrepreneurship Cell
Seats: 75
Learn how to convert an idea into a startup through problem identification, validation and pitching.

₹100

Register

6.3 :- Search



StudentHub Portal

Empowering Students Through Technology

[Home](#)
[About](#)
[Register](#)
[Login](#)
[Dashboard](#)
[Events](#)
[Profile](#)
[Contact](#)
[Admin](#)
[FAQ](#)
[Feedback](#)

Upcoming Events

Explore coding competitions, hackathons, workshops, expert talks, seminars and cultural activities happening on campus.

Hackathon

Hackathon 2026

Date: 3 October 2026
Time: 8:00 AM - 6:00 PM
Venue: Innovation Lab
Organizer: Innovation Club
Seats: 120
Build innovative technology solutions for real-world problems in a team-based hackathon.

₹100

Register

Hackathon

Innovation Hack Challenge

Date: 20 November 2026
Time: 8:00 AM - 5:00 PM
Venue: Innovation Lab
Organizer: Innovation Club
Seats: 100
Develop creative technology solutions and present your working prototype to a judging panel.

₹100

Register

[Previous](#)
Page 1 of 1
[Next](#)

[Home](#)
[About](#)
[Register](#)
[Login](#)
[Dashboard](#)
[Events](#)
[Profile](#)
[Contact](#)
[Admin](#)
[FAQ](#)
[Feedback](#)

Upcoming Events

Explore coding competitions, hackathons, workshops, expert talks, seminars and cultural activities happening on campus.

Workshop

Web Development Workshop

Date: 7 October 2026
Time: 10:00 AM - 1:00 PM
Venue: Web Technology Lab
Organizer: Web Development Club
Seats: 80
Learn HTML, CSS, responsive design and modern web development through practical activities.

Free

Register

Workshop

Startup & Innovation Bootcamp

Date: 22 October 2026
Time: 10:00 AM - 4:00 PM
Venue: Innovation Center
Organizer: Entrepreneurship Cell
Seats: 75
Learn how to convert an idea into a startup through problem identification, validation and pitching.

₹100

Register

Workshop

Git & GitHub Hands-on Workshop

Date: 30 October 2026
Time: 10:00 AM - 1:00 PM
Venue: Computer Laboratory
Organizer: Developer Student Club
Seats: 90
Learn version control, Git commands, GitHub repositories, branching and collaborative development.

₹100

Register

Workshop

Generative AI Workshop

Date: 12 November 2026
Time: 10:00 AM - 2:00 PM
Venue: AI Laboratory
Organizer: AI Club
Seats: 80
Explore Generative AI concepts, prompt engineering and practical applications of modern AI tools.

₹100

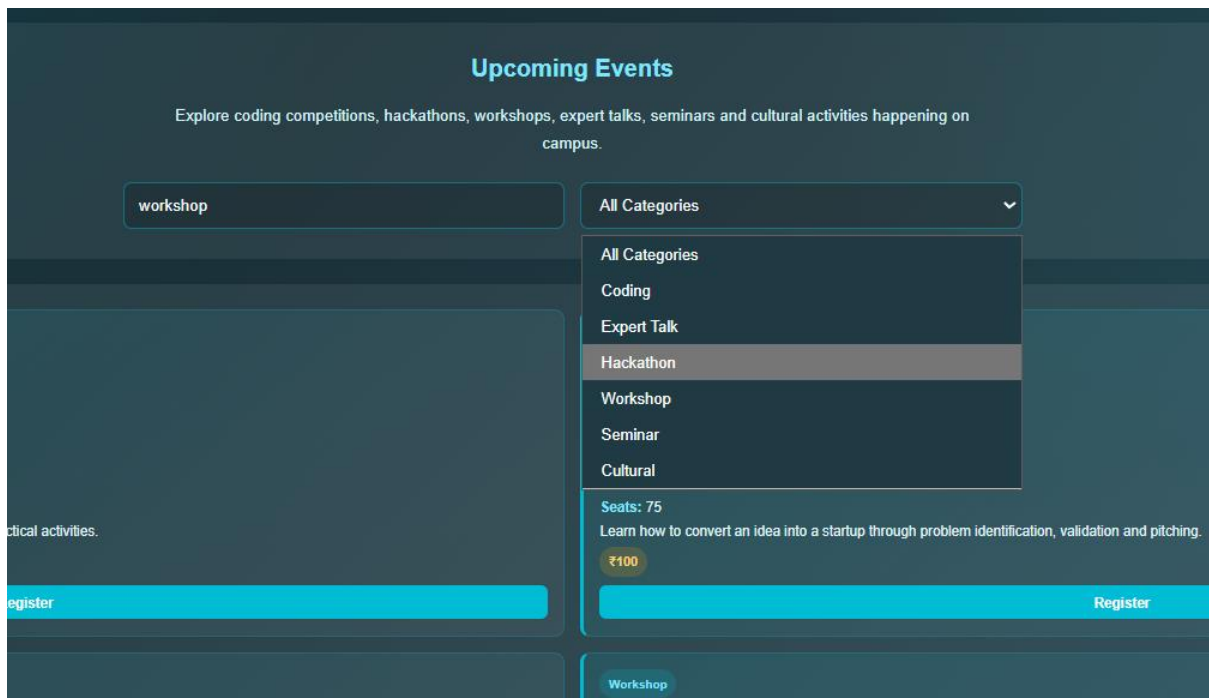
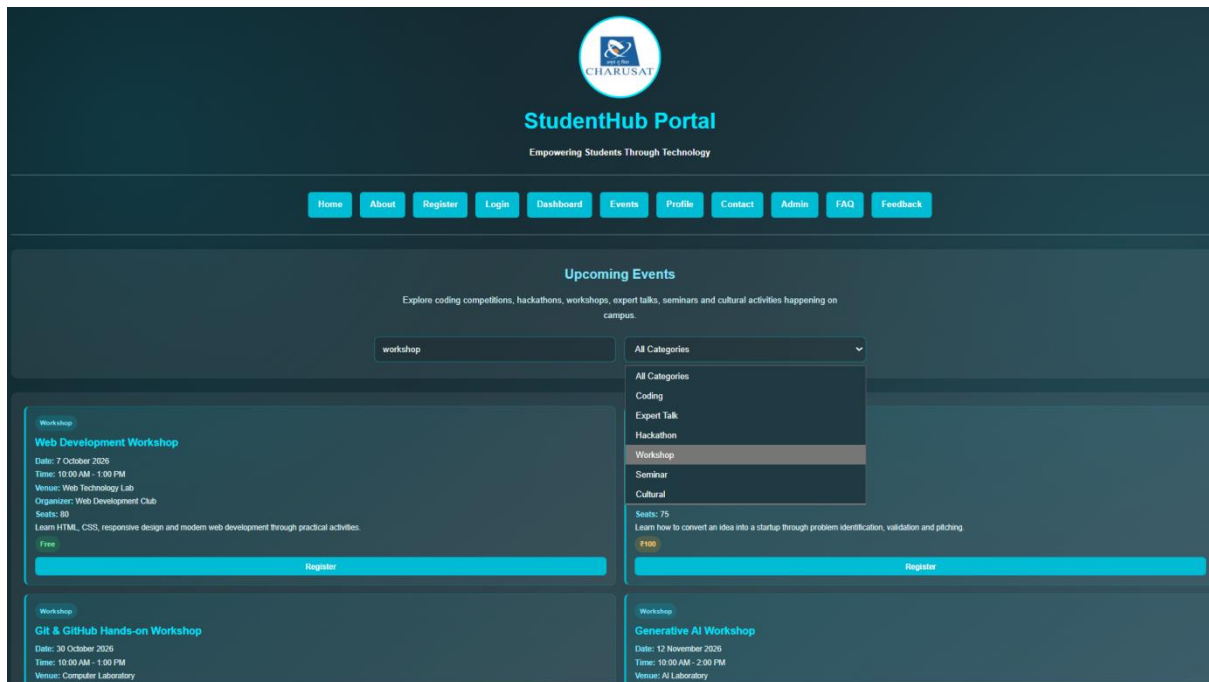
Register

Workshop

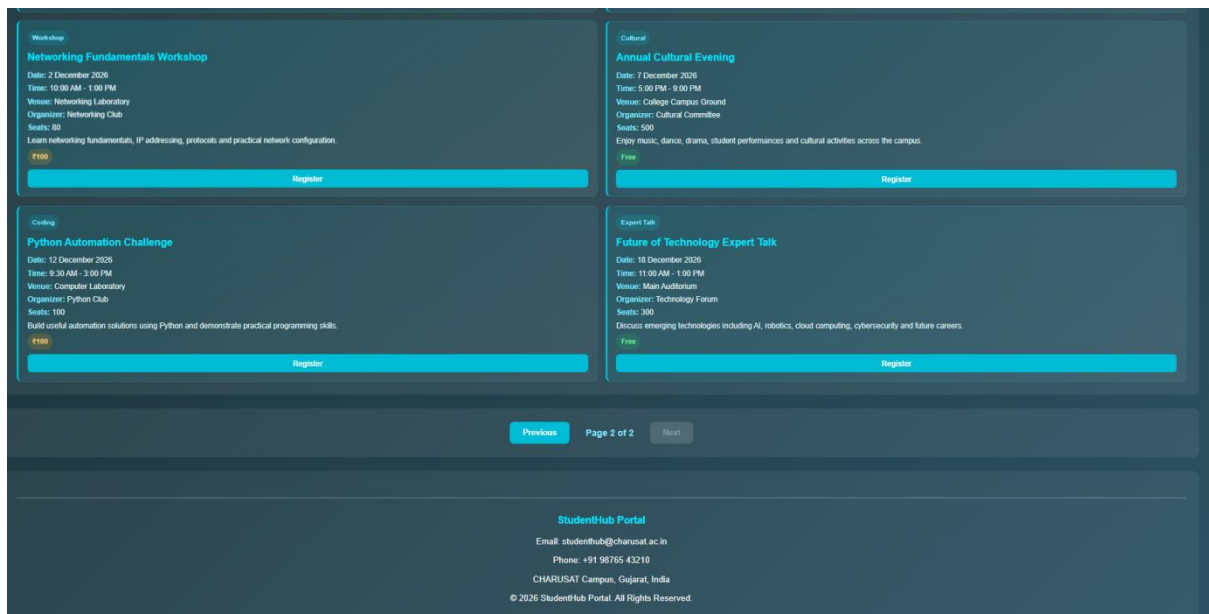
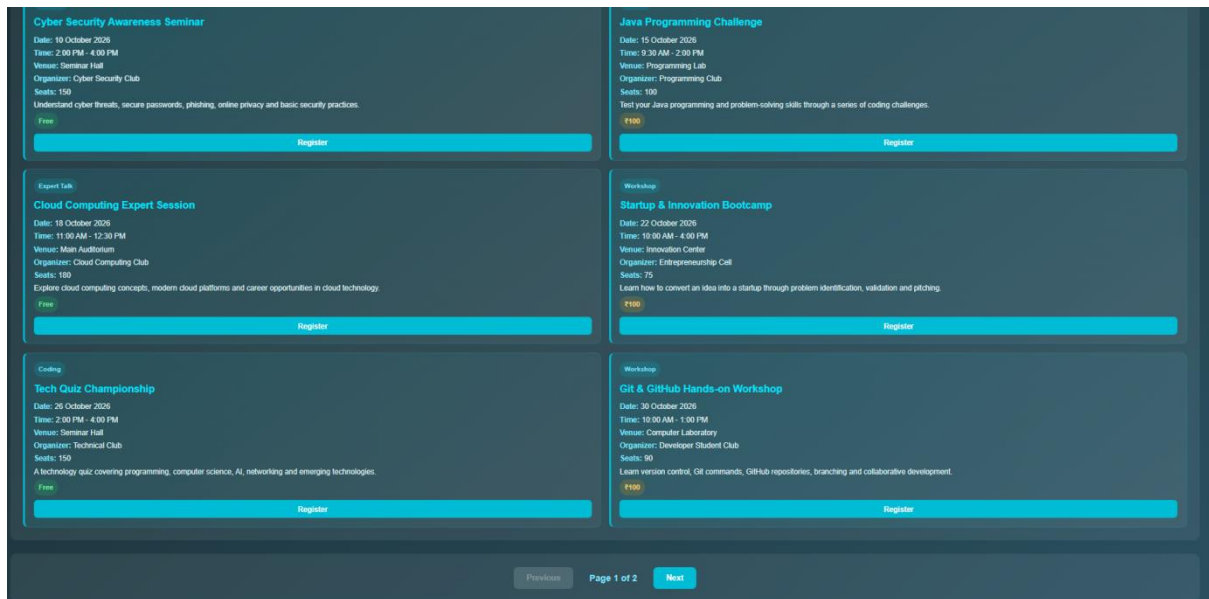
Networking Fundamentals Workshop

Date: 2 December 2026

6.4 :- Category Filter



6.5 :- Pagination



11. Result

The external JSON data was successfully fetched using the Fetch API and dynamically rendered on the StudentHub webpage. Search, category filtering, and pagination were successfully implemented using JavaScript array and DOM manipulation techniques.

Learning Outcomes

After completing this practical, I was able to:

- Understand the structure and use of JSON data.
- Fetch external data using the Fetch API.
- Parse JSON responses and use them in JavaScript.
- Dynamically render data on an HTML webpage.
- Use JavaScript array methods such as `map()`, `filter()`, `sort()`, and `slice()`.
- Implement search and filtering functionality.
- Understand and implement pagination logic.
- Handle basic loading and error conditions while fetching data.

Conclusion

Thus, JSON data was successfully fetched using the Fetch API and dynamically displayed on the StudentHub Events page. JavaScript array methods were used to implement search, filtering, and pagination. This practical helped in understanding how modern web applications consume external data and dynamically update the user interface.

Faculty Remarks :- _____